HOMEWORK 2.3 · UNIT 2 · NC.8.EE.1

Power-to-a-Power; Negative & Zero Exponents

Sep 2 / Sep 3 · Targets: EE.1-2, EE.1-3, EE.1-7, EE.1-6, EE.1-4, EE.1-8, EE.1-9

Name _____ Date _____ Block _____

Remember: Power to a power → multiply: $(x^2)^3 = x^6$. Raise every factor: $(3x)^2 = 9x^2$. **Zero:** any nonzero base$^0 = 1$.
Negative: flip to a fraction: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$, $5x^{-2} = \frac{5}{x^2}$.

SCORE 3.0 Approaching · the basics

Targets: EE.1-2, EE.1-3

1 Simplify. Write as one power. $(5^2)^3$ EE.1-2

2 Simplify. $(y^5)^4$ EE.1-2

3 Find the value. 4^{-2} EE.1-3

4 What is the value of $(-6)^0$? EE.1-3

Ⓐ −6 Ⓑ 0 Ⓒ 1 Ⓓ −1

5 Simplify. $\frac{x^9}{x^2}$ REVIEW 2.2 EE.1-2

SCORE 4.0 Proficient · on target

Targets: EE.1-7, EE.1-6, EE.1-4

6 **Lou** made a mistake. Circle it. Explain it. Fix it. EE.1-7

LOU'S WORK
$(x^3)^4 = x^7$

7 Simplify. Use positive exponents. $(2x^{-3})^2$ EE.1-6

Hint: Square the 2 too.

8 Simplify. $(2a^4)(7a^3)$ REVIEW 2.2 EE.1-4

SCORE 5.0 Mastery · go deeper

Targets: EE.1-8, EE.1-9

9 Use the pattern. Explain why $2^{-2} = \frac{1}{4}$. EE.1-8

Power	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
Value	8	4	2	?	?	?

Each time the exponent goes down 1, you _____. So 2^{-1} = _____ and 2^{-2} = _____.

10 Kim says $3x^{-2} = \frac{1}{3x^2}$. Do you agree? Explain. EE.1-9

I _____ because the exponent is only on _____.